DBMS LAB · ASSIGNMENT 6

Implementation of Data Integrity Constraints

Keys, UNIQUE, CHECK, DEFAULT and triggers in the airline database, with test cases

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PROJECT
Airline Reservation & Flight Booking System

DATABASE
MySQL 8.0 · `airline_reservation`

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Aim

To enforce the rules of the airline inside the database: a seat can be sold only once, a cabin cannot be oversold, a cancelled flight cannot be booked, and a booking cannot be paid twice. This assignment explains why each constraint is needed and proves the rules with test cases.

1. Constraint Plan

Rule type	Mechanism	Count
Every row identifiable	PRIMARY KEY (natural key for airports and classes)	9
Links between tables are valid	FOREIGN KEY with RESTRICT / CASCADE	10
Natural identifiers and one-of-a-kind facts	UNIQUE (4 on generated columns)	11
Values within real-world limits	CHECK, ENUM, NOT NULL	19 CHECK
Sensible values when omitted	DEFAULT	13
Rules that need other tables	BEFORE / AFTER triggers	3

2. Primary and Foreign Keys

```
SELECT TABLE_NAME,
       SUM(CONSTRAINT_TYPE = 'PRIMARY KEY') AS pk,
       SUM(CONSTRAINT_TYPE = 'FOREIGN KEY') AS fks,
       GROUP_CONCAT(IF(CONSTRAINT_TYPE = 'FOREIGN KEY', CONSTRAINT_NAME, NULL)) AS fk_names
FROM information_schema.TABLE_CONSTRAINTS
WHERE TABLE_SCHEMA = 'airline_reservation'
GROUP BY TABLE_NAME
ORDER BY fks, TABLE_NAME;
```

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```
mysql> SELECT TABLE_NAME,
    ->        SUM(CONSTRAINT_TYPE = 'PRIMARY KEY') AS pk,
    ->        SUM(CONSTRAINT_TYPE = 'FOREIGN KEY') AS fks,
    ->        GROUP_CONCAT(IF(CONSTRAINT_TYPE = 'FOREIGN KEY', CONSTRAINT_NAME, NULL)) AS fk_names
    -> FROM information_schema.TABLE_CONSTRAINTS
    -> WHERE TABLE_SCHEMA = 'airline_reservation'
    -> GROUP BY TABLE_NAME
    -> ORDER BY fks, TABLE_NAME;
+----------------+------+------+-------------------------------------------------------------------------+
| TABLE_NAME     | pk   | fks  | fk_names                                                                |
+----------------+------+------+-------------------------------------------------------------------------+
| aircraft       |    1 |    0 | NULL                                                                    |
| airports       |    1 |    0 | NULL                                                                    |
| passengers     |    1 |    0 | NULL                                                                    |
| travel_classes |    1 |    0 | NULL                                                                    |
| bookings       |    1 |    1 | fk_booking_passenger                                                    |
| payments       |    1 |    1 | fk_payment_booking                                                      |
| flights        |    1 |    2 | fk_flight_aircraft,fk_flight_route                                      |
| routes         |    1 |    2 | fk_route_destination,fk_route_origin                                    |
| ticket_details |    1 |    4 | fk_ticket_booking,fk_ticket_class,fk_ticket_flight,fk_ticket_passenger |
+----------------+------+------+-------------------------------------------------------------------------+
9 rows in set (0.00 sec)
```

- **airports** uses the IATA code and **travel_classes** uses a two-letter class code as natural primary keys. Both are short, stable and meaningful. All other entities use AUTO_INCREMENT surrogate keys.
- **ticket_details** has four foreign keys (booking, flight, traveller, class), so a ticket can only exist for real rows in all four parent tables.
- The two airport foreign keys on **routes** use ON UPDATE RESTRICT. MySQL does not allow a CHECK constraint on a column that a cascading foreign key may change, and these columns are needed by the CHECK origin ≠ destination. IATA codes never change anyway.

3. UNIQUE constraints

Besides natural identifiers, four rules use **generated columns**. A generated column holds a value only while the row is “live” and NULL otherwise, and a UNIQUE index allows many NULLs. This gives one flight per route per day, one live ticket per seat, one live ticket per traveller per flight, and one successful payment per booking.

```
ALTER TABLE aircraft       ADD CONSTRAINT uk_aircraft_registration UNIQUE (registration);
ALTER TABLE travel_classes ADD CONSTRAINT uk_class_name            UNIQUE (class_name);
ALTER TABLE routes         ADD CONSTRAINT uk_route_flight_number   UNIQUE (flight_number);
ALTER TABLE passengers     ADD CONSTRAINT uk_passenger_email       UNIQUE (email);
ALTER TABLE passengers     ADD CONSTRAINT uk_passenger_passport    UNIQUE (nationality, passport_number);
ALTER TABLE bookings       ADD CONSTRAINT uk_booking_reference     UNIQUE (booking_reference);
ALTER TABLE payments       ADD CONSTRAINT uk_payment_txn_ref       UNIQUE (transaction_ref);

-- A route operates at most once per day.
ALTER TABLE flights
    ADD COLUMN departure_date DATE GENERATED ALWAYS AS (DATE(departure_time)) VIRTUAL,
    ADD CONSTRAINT uk_flight_route_day UNIQUE (route_id, departure_date);

-- A seat can be held by only one live ticket, and a traveller can hold only
-- one live ticket on a flight. Cancelled tickets get NULL and stop competing.
ALTER TABLE ticket_details
    ADD COLUMN active_seat VARCHAR(4)
        GENERATED ALWAYS AS (IF(ticket_status = 'Cancelled', NULL, seat_number)) VIRTUAL,
    ADD COLUMN active_traveller INT
        GENERATED ALWAYS AS (IF(ticket_status = 'Cancelled', NULL, passenger_id)) VIRTUAL,
    ADD CONSTRAINT uk_ticket_seat      UNIQUE (flight_id, active_seat),
    ADD CONSTRAINT uk_ticket_traveller UNIQUE (flight_id, active_traveller);

-- At most one successful payment per booking (failed attempts are allowed).
ALTER TABLE payments
    ADD COLUMN success_booking INT
        GENERATED ALWAYS AS (IF(payment_status = 'Success', booking_id, NULL)) VIRTUAL,
    ADD CONSTRAINT uk_one_successful_payment UNIQUE (success_booking);
```

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```
mysql> ALTER TABLE aircraft       ADD CONSTRAINT uk_aircraft_registration UNIQUE (registration);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE travel_classes ADD CONSTRAINT uk_class_name            UNIQUE (class_name);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE routes         ADD CONSTRAINT uk_route_flight_number   UNIQUE (flight_number);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE passengers     ADD CONSTRAINT uk_passenger_email       UNIQUE (email);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE passengers     ADD CONSTRAINT uk_passenger_passport    UNIQUE (nationality, passport_number
);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE bookings       ADD CONSTRAINT uk_booking_reference     UNIQUE (booking_reference);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE payments       ADD CONSTRAINT uk_payment_txn_ref       UNIQUE (transaction_ref);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE flights
    ->     ADD COLUMN departure_date DATE GENERATED ALWAYS AS (DATE(departure_time)) VIRTUAL,
    ->     ADD CONSTRAINT uk_flight_route_day UNIQUE (route_id, departure_date);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE ticket_details
    ->     ADD COLUMN active_seat VARCHAR(4)
    ->         GENERATED ALWAYS AS (IF(ticket_status = 'Cancelled', NULL, seat_number)) VIRTUAL,
    ->     ADD COLUMN active_traveller INT
    ->         GENERATED ALWAYS AS (IF(ticket_status = 'Cancelled', NULL, passenger_id)) VIRTUAL,
    ->     ADD CONSTRAINT uk_ticket_seat      UNIQUE (flight_id, active_seat),
    ->     ADD CONSTRAINT uk_ticket_traveller UNIQUE (flight_id, active_traveller);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE payments
    ->     ADD COLUMN success_booking INT
    ->         GENERATED ALWAYS AS (IF(payment_status = 'Success', booking_id, NULL)) VIRTUAL,
    ->     ADD CONSTRAINT uk_one_successful_payment UNIQUE (success_booking);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

4. CHECK constraints

CHECK constraints validate codes and formats and compare columns of the same row (origin ≠ destination, arrival > departure).

```sql
ALTER TABLE airports
    ADD CONSTRAINT ck_airport_code CHECK (airport_code REGEXP '^[A-Z]{3}$');

ALTER TABLE aircraft
    ADD CONSTRAINT ck_aircraft_seats CHECK (economy_seats > 0 AND premium_seats >= 0
                                        AND business_seats >= 0 AND first_seats >= 0),
    ADD CONSTRAINT ck_aircraft_registration CHECK (registration REGEXP '^VT-[A-Z]{3}$');

ALTER TABLE travel_classes
    ADD CONSTRAINT ck_class_multiplier CHECK (fare_multiplier BETWEEN 1.00 AND 10.00),
    ADD CONSTRAINT ck_class_baggage    CHECK (checked_baggage_kg BETWEEN 0 AND 60);

ALTER TABLE routes
    ADD CONSTRAINT ck_route_airports CHECK (origin_code <> destination_code),
    ADD CONSTRAINT ck_route_duration CHECK (duration_min BETWEEN 20 AND 1200),
    ADD CONSTRAINT ck_route_distance CHECK (distance_km > 0),
    ADD CONSTRAINT ck_route_flight_no CHECK (flight_number REGEXP '^[A-Z]{2}[0-9]{2,4}$');

ALTER TABLE flights
    ADD CONSTRAINT ck_flight_times CHECK (arrival_time > departure_time),
    ADD CONSTRAINT ck_flight_price CHECK (base_price > 0);

ALTER TABLE passengers
    ADD CONSTRAINT ck_passenger_email    CHECK (email REGEXP '^[^@ ]+@[^@ ]+\\.[a-z]{2,}$'),
    ADD CONSTRAINT ck_passenger_phone    CHECK (phone REGEXP '^\\+?[0-9]{10,13}$'),
    ADD CONSTRAINT ck_passenger_passport CHECK (passport_number REGEXP '^[A-Z0-9]{6,9}$'),
    ADD CONSTRAINT ck_passenger_dob      CHECK (date_of_birth >= '1900-01-01');

ALTER TABLE bookings
    ADD CONSTRAINT ck_booking_reference CHECK (booking_reference REGEXP '^[A-Z0-9]{6}$');

ALTER TABLE ticket_details
    ADD CONSTRAINT ck_ticket_seat CHECK (seat_number REGEXP '^[0-9]{1,2}[A-K]$'),
    ADD CONSTRAINT ck_ticket_fare CHECK (fare > 0);

ALTER TABLE payments
    ADD CONSTRAINT ck_payment_amount CHECK (amount > 0);
```

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```
mysql> ALTER TABLE airports
    ->     ADD CONSTRAINT ck_airport_code CHECK (airport_code REGEXP '^[A-Z]{3}$');
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE aircraft
    ->     ADD CONSTRAINT ck_aircraft_seats CHECK (economy_seats > 0 AND premium_seats >= 0
    ->                                             AND business_seats >= 0 AND first_seats >= 0),
    ->     ADD CONSTRAINT ck_aircraft_registration CHECK (registration REGEXP '^VT-[A-Z]{3}$');
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE travel_classes
    ->     ADD CONSTRAINT ck_class_multiplier CHECK (fare_multiplier BETWEEN 1.00 AND 10.00),
    ->     ADD CONSTRAINT ck_class_baggage    CHECK (checked_baggage_kg BETWEEN 0 AND 60);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE routes
    ->     ADD CONSTRAINT ck_route_airports CHECK (origin_code <> destination_code),
    ->     ADD CONSTRAINT ck_route_duration CHECK (duration_min BETWEEN 20 AND 1200),
    ->     ADD CONSTRAINT ck_route_distance CHECK (distance_km > 0),
    ->     ADD CONSTRAINT ck_route_flight_no CHECK (flight_number REGEXP '^[A-Z]{2}[0-9]{2,4}$');
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE flights
    ->     ADD CONSTRAINT ck_flight_times CHECK (arrival_time > departure_time),
    ->     ADD CONSTRAINT ck_flight_price CHECK (base_price > 0);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE passengers
    ->     ADD CONSTRAINT ck_passenger_email    CHECK (email REGEXP '^[^@ ]+@[^@ ]+\\.[a-z]{2,}$'),
    ->     ADD CONSTRAINT ck_passenger_phone    CHECK (phone REGEXP '^\\+?[0-9]{10,13}$'),
    ->     ADD CONSTRAINT ck_passenger_passport CHECK (passport_number REGEXP '^[A-Z0-9]{6,9}$'),
    ->     ADD CONSTRAINT ck_passenger_dob      CHECK (date_of_birth >= '1900-01-01');
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE bookings
    ->     ADD CONSTRAINT ck_booking_reference CHECK (booking_reference REGEXP '^[A-Z0-9]{6}$');
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE ticket_details
    ->     ADD CONSTRAINT ck_ticket_seat CHECK (seat_number REGEXP '^[0-9]{1,2}[A-K]$'),
    ->     ADD CONSTRAINT ck_ticket_fare CHECK (fare > 0);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE payments
    ->     ADD CONSTRAINT ck_payment_amount CHECK (amount > 0);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

5. DEFAULT values

Defaults describe the normal case: a new flight is Scheduled, a new booking is Pending, a ticket is Economy and Booked, a payment is Pending, and timestamps are filled in automatically.

```sql
ALTER TABLE airports       ALTER country          SET DEFAULT 'India';
ALTER TABLE aircraft       ALTER premium_seats    SET DEFAULT 0;
ALTER TABLE aircraft       ALTER business_seats   SET DEFAULT 0;
ALTER TABLE aircraft       ALTER first_seats      SET DEFAULT 0;
ALTER TABLE aircraft       ALTER status           SET DEFAULT 'Active';
ALTER TABLE flights        ALTER flight_status    SET DEFAULT 'Scheduled';
ALTER TABLE passengers     ALTER nationality      SET DEFAULT 'Indian';
ALTER TABLE bookings       ALTER booking_date     SET DEFAULT (CURRENT_TIMESTAMP);
ALTER TABLE bookings       ALTER status           SET DEFAULT 'Pending';
ALTER TABLE ticket_details ALTER class_code       SET DEFAULT 'EC';
ALTER TABLE ticket_details ALTER ticket_status    SET DEFAULT 'Booked';
ALTER TABLE payments       ALTER payment_status   SET DEFAULT 'Pending';
ALTER TABLE payments       ALTER transaction_date SET DEFAULT (CURRENT_TIMESTAMP);
```

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```
mysql> ALTER TABLE airports       ALTER country          SET DEFAULT 'India';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE aircraft       ALTER premium_seats    SET DEFAULT 0;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE aircraft       ALTER business_seats   SET DEFAULT 0;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE aircraft       ALTER first_seats      SET DEFAULT 0;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE aircraft       ALTER status           SET DEFAULT 'Active';
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE flights        ALTER flight_status    SET DEFAULT 'Scheduled';
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE passengers     ALTER nationality      SET DEFAULT 'Indian';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE bookings       ALTER booking_date     SET DEFAULT (CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE bookings       ALTER status           SET DEFAULT 'Pending';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE ticket_details ALTER class_code       SET DEFAULT 'EC';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE ticket_details ALTER ticket_status    SET DEFAULT 'Booked';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE payments       ALTER payment_status   SET DEFAULT 'Pending';
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE payments       ALTER transaction_date SET DEFAULT (CURRENT_TIMESTAMP);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

6. Triggers

Three rules need data from other tables, so they are triggers: the flight must still be open and the cabin must have a seat (and the fare is computed); a cancelled flight cancels its tickets; and a cancelled booking cannot be paid.

```
DELIMITER //

-- Selling a seat: the flight must still be open for sale, the cabin must
-- have a free seat, and the fare defaults to base price x class multiplier.
CREATE TRIGGER trg_ticket_before_insert
BEFORE INSERT ON ticket_details
FOR EACH ROW
BEGIN
    DECLARE v_status     VARCHAR(10);
    DECLARE v_price      DECIMAL(10,2);
    DECLARE v_capacity   SMALLINT;
    DECLARE v_sold       INT;
    DECLARE v_multiplier DECIMAL(4,2);

    SELECT f.flight_status, f.base_price,
           CASE NEW.class_code
               WHEN 'EC' THEN a.economy_seats  WHEN 'PE' THEN a.premium_seats
               WHEN 'BU' THEN a.business_seats WHEN 'FC' THEN a.first_seats END
      INTO v_status, v_price, v_capacity
      FROM flights f JOIN aircraft a ON a.aircraft_id = f.aircraft_id
     WHERE f.flight_id = NEW.flight_id;

    IF v_status NOT IN ('Scheduled', 'Delayed') THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Ticket refused: flight is no longer open for booking';
    END IF;

    SELECT COUNT(*) INTO v_sold FROM ticket_details
     WHERE flight_id = NEW.flight_id AND class_code = NEW.class_code
       AND ticket_status <> 'Cancelled';
    IF v_sold >= v_capacity THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Ticket refused: no seats left in this travel class';
    END IF;

    IF NEW.fare IS NULL THEN
        SELECT fare_multiplier INTO v_multiplier
          FROM travel_classes WHERE class_code = NEW.class_code;
        SET NEW.fare = ROUND(v_price * v_multiplier, 2);
    END IF;
END//

-- Cancelling a flight cancels every live ticket on it.
CREATE TRIGGER trg_flight_after_update
AFTER UPDATE ON flights
FOR EACH ROW
BEGIN
    IF NEW.flight_status = 'Cancelled' AND OLD.flight_status <> 'Cancelled' THEN
        UPDATE ticket_details SET ticket_status = 'Cancelled'
         WHERE flight_id = NEW.flight_id AND ticket_status <> 'Cancelled';
    END IF;
END//

-- A cancelled booking cannot receive a new pending or successful payment.
CREATE TRIGGER trg_payment_before_insert
BEFORE INSERT ON payments
FOR EACH ROW
BEGIN
    IF NEW.payment_status IN ('Pending', 'Success') AND
       (SELECT status FROM bookings WHERE booking_id = NEW.booking_id) = 'Cancelled' THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Payment refused: booking is cancelled';
    END IF;
END//

DELIMITER ;
```

```
SELECT TRIGGER_NAME, ACTION_TIMING AS timing, EVENT_MANIPULATION AS event, EVENT_OBJECT_TABLE AS on_table
FROM information_schema.TRIGGERS WHERE TRIGGER_SCHEMA = 'airline_reservation';
```

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```
mysql> SELECT TRIGGER_NAME, ACTION_TIMING AS timing, EVENT_MANIPULATION AS event, EVENT_OBJECT_TABLE AS on_tab
le
    -> FROM information_schema.TRIGGERS WHERE TRIGGER_SCHEMA = 'airline_reservation';
+---------------------------+--------+--------+----------------+
| TRIGGER_NAME              | timing | event  | on_table       |
+---------------------------+--------+--------+----------------+
| trg_ticket_before_insert  | BEFORE | INSERT | ticket_details |
| trg_flight_after_update   | AFTER  | UPDATE | flights        |
| trg_payment_before_insert | BEFORE | INSERT | payments       |
+---------------------------+--------+--------+----------------+
3 rows in set (0.00 sec)
```

7. Purpose of Each Constraint

Constraint	Purpose
uk_aircraft_registration, ck_aircraft_registration	Each aircraft has one Indian registration (VT-XXX).
uk_route_flight_number, ck_route_flight_no	A flight number (two letters + digits) names exactly one route.
uk_flight_route_day	The same route cannot be scheduled twice on one day.
uk_passenger_email	One customer account per e-mail address.
uk_passenger_passport, ck_passenger_passport	A passport is unique within its issuing country; format A–Z/0–9, 6–9 characters.
uk_booking_reference, ck_booking_reference	Six-character PNR, unique across all bookings.
uk_ticket_seat	A seat on a flight can be held by only one live ticket (no double-booking).
uk_ticket_traveller	A traveller cannot occupy two seats on the same flight.
uk_one_successful_payment, uk_payment_txn_ref	A booking is paid once; every gateway transaction is recorded once.
ck_airport_code	IATA format: three capital letters.
ck_aircraft_seats	At least one economy seat; no negative seat counts.
ck_class_multiplier, ck_class_baggage	Realistic fare multipliers (1–10×) and baggage allowances (0–60 kg).
ck_route_airports / duration / distance	A route connects two different airports with a plausible duration and distance.
ck_flight_times, ck_flight_price	Arrival after departure; positive base price.
ck_passenger_email / phone / dob	Valid e-mail and international phone format; plausible birth date.
ck_ticket_seat, ck_ticket_fare	Seat numbers such as 12A; positive fare.
ck_payment_amount	Payments are for a positive amount.
DEFAULTs	country India, nationality Indian, seat counts 0, aircraft Active, flight Scheduled, booking Pending + timestamp, ticket Economy + Booked, payment Pending + timestamp.
trg_ticket_before_insert	Refuses tickets on flights that are not open and on full cabins; computes the fare.
trg_flight_after_update	Cancelling a flight cancels all its live tickets in the same statement.
trg_payment_before_insert	Stops pending or successful payments on cancelled bookings.

8. Test Cases

The tests run on the empty database before the Assignment 7 data is loaded. A small fixture is inserted first, using a test aircraft with only two economy seats and one business seat so that capacity limits are easy to reach. Rejected statements change nothing. Destructive tests are rolled back.

8.1 Fixture

Airports, a tiny test aircraft (2 economy seats, 1 business seat) and two travel classes

```
INSERT INTO airports (airport_code, airport_name, city) VALUES
    ('HYD', 'Rajiv Gandhi International Airport', 'Hyderabad'),
    ('DEL', 'Indira Gandhi International Airport', 'New Delhi');
INSERT INTO aircraft (aircraft_id, registration, model, economy_seats, business_seats, in_service_since)
VALUES (1, 'VT-TST', 'Test Trainer', 2, 1, '2024-01-01');
INSERT INTO travel_classes VALUES ('EC', 'Economy', 1.00, 15), ('BU', 'Business', 3.20, 35);
SELECT airport_code, city, country FROM airports;
SELECT aircraft_id, registration, economy_seats, premium_seats, business_seats, first_seats, status FROM aircraft;
```

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```
mysql> INSERT INTO airports (airport_code, airport_name, city) VALUES
    ->     ('HYD', 'Rajiv Gandhi International Airport', 'Hyderabad'),
    ->     ('DEL', 'Indira Gandhi International Airport', 'New Delhi');
Query OK, 2 rows affected (0.04 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO aircraft (aircraft_id, registration, model, economy_seats, business_seats, in_service_since
)
    -> VALUES (1, 'VT-TST', 'Test Trainer', 2, 1, '2024-01-01');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO travel_classes VALUES ('EC', 'Economy', 1.00, 15), ('BU', 'Business', 3.20, 35);
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT airport_code, city, country FROM airports;
+--------------+-----------+---------+
| airport_code | city      | country |
+--------------+-----------+---------+
| DEL          | New Delhi | India   |
| HYD          | Hyderabad | India   |
+--------------+-----------+---------+
2 rows in set (0.00 sec)

mysql> SELECT aircraft_id, registration, economy_seats, premium_seats, business_seats, first_seats, status FRO
M aircraft;
+-------------+--------------+---------------+---------------+----------------+-------------+--------+
| aircraft_id | registration | economy_seats | premium_seats | business_seats | first_seats | status |
+-------------+--------------+---------------+---------------+----------------+-------------+--------+
|           1 | VT-TST       |             2 |             0 |              1 |           0 | Active |
+-------------+--------------+---------------+---------------+----------------+-------------+--------+
1 row in set (0.00 sec)
```

A route and its flight on 1 December 2026 (flight_status uses DEFAULT)

```
INSERT INTO routes (route_id, flight_number, origin_code, destination_code, scheduled_departure, duration_min, distance_km)
VALUES (1, 'TS101', 'HYD', 'DEL', '06:00', 130, 1265);
INSERT INTO flights (flight_id, route_id, aircraft_id, departure_time, arrival_time, base_price)
VALUES (1, 1, 1, '2026-12-01 06:00', '2026-12-01 08:10', 5000.00);
SELECT flight_id, route_id, departure_time, departure_date, base_price, flight_status FROM flights;
```

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```
mysql> INSERT INTO routes (route_id, flight_number, origin_code, destination_code, scheduled_departure, durati
on_min, distance_km)
    -> VALUES (1, 'TS101', 'HYD', 'DEL', '06:00', 130, 1265);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO flights (flight_id, route_id, aircraft_id, departure_time, arrival_time, base_price)
    -> VALUES (1, 1, 1, '2026-12-01 06:00', '2026-12-01 08:10', 5000.00);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT flight_id, route_id, departure_time, departure_date, base_price, flight_status FROM flights;
+-----------+----------+---------------------+----------------+------------+---------------+
| flight_id | route_id | departure_time      | departure_date | base_price | flight_status |
+-----------+----------+---------------------+----------------+------------+---------------+
|         1 |        1 | 2026-12-01 06:00:00 | 2026-12-01     |    5000.00 | Scheduled     |
+-----------+----------+---------------------+----------------+------------+---------------+
1 row in set (0.00 sec)
```

Passengers (nationality uses DEFAULT) and a booking (booking_date and status use DEFAULT)

```
INSERT INTO passengers (passenger_id, first_name, last_name, email, phone, passport_number, date_of_birth) VALUES
    (1, 'Test',   'Traveller', 'test.traveller@example.com', '+919876543210', 'M1234567', '1995-05-05'),
    (2, 'Second', 'Traveller', 'second.trav@example.com',   '+919123456780', 'N7654321', '1990-01-01'),
    (3, 'Third',  'Traveller', 'third.trav@example.com',    '+919000000003', 'P1112223', '2010-09-09');
INSERT INTO bookings (booking_id, passenger_id, booking_reference) VALUES (1, 1, 'TEST01');
SELECT passenger_id, first_name, nationality FROM passengers;
SELECT booking_id, passenger_id, booking_reference, booking_date, status FROM bookings;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO passengers (passenger_id, first_name, last_name, email, phone, passport_number, date_of_bir
th) VALUES
    ->     (1, 'Test',   'Traveller', 'test.traveller@example.com', '+919876543210', 'M1234567', '1995-05-05')
,
    ->     (2, 'Second', 'Traveller', 'second.trav@example.com',   '+919123456780', 'N7654321', '1990-01-01'),
    ->     (3, 'Third',  'Traveller', 'third.trav@example.com',    '+919000000003', 'P1112223', '2010-09-09');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO bookings (booking_id, passenger_id, booking_reference) VALUES (1, 1, 'TEST01');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT passenger_id, first_name, nationality FROM passengers;
+--------------+------------+-------------+
| passenger_id | first_name | nationality |
+--------------+------------+-------------+
|            1 | Test       | Indian      |
|            2 | Second     | Indian      |
|            3 | Third      | Indian      |
+--------------+------------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT booking_id, passenger_id, booking_reference, booking_date, status FROM bookings;
+------------+--------------+-------------------+---------------------+---------+
| booking_id | passenger_id | booking_reference | booking_date        | status  |
+------------+--------------+-------------------+---------------------+---------+
|          1 |            1 | TEST01            | 2026-10-25 15:35:31 | Pending |
+------------+--------------+-------------------+---------------------+---------+
1 row in set (0.00 sec)
```

8.2 Results

TC-01 Primary key: an airport code can exist only once

```
INSERT INTO airports (airport_code, airport_name, city) VALUES ('HYD', 'Duplicate', 'Hyderabad');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO airports (airport_code, airport_name, city) VALUES ('HYD', 'Duplicate', 'Hyderabad');
ERROR 1062 (23000): Duplicate entry 'HYD' for key 'airports.PRIMARY'
```

TC-02 NOT NULL: e-mail is mandatory

```
INSERT INTO passengers (first_name, last_name, phone, passport_number, date_of_birth)
VALUES ('No', 'Email', '+919000000009', 'Z9999999', '2000-01-01');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO passengers (first_name, last_name, phone, passport_number, date_of_birth)
    -> VALUES ('No', 'Email', '+919000000009', 'Z9999999', '2000-01-01');
ERROR 1364 (HY000): Field 'email' doesn't have a default value
```

TC-03 Two passengers cannot share an e-mail

```
INSERT INTO passengers (first_name, last_name, email, phone, passport_number, date_of_birth)
VALUES ('Copy', 'Cat', 'test.traveller@example.com', '+919000000004', 'K4445556', '1999-01-01');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO passengers (first_name, last_name, email, phone, passport_number, date_of_birth)
    -> VALUES ('Copy', 'Cat', 'test.traveller@example.com', '+919000000004', 'K4445556', '1999-01-01');
ERROR 1062 (23000): Duplicate entry 'test.traveller@example.com' for key 'passengers.uk_passenger_email'
```

TC-04 Same passport number and same nationality is rejected

```
INSERT INTO passengers (first_name, last_name, email, phone, passport_number, date_of_birth)
VALUES ('Fake', 'Passport', 'fake@example.com', '+919000000005', 'M1234567', '1980-01-01');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO passengers (first_name, last_name, email, phone, passport_number, date_of_birth)
    -> VALUES ('Fake', 'Passport', 'fake@example.com', '+919000000005', 'M1234567', '1980-01-01');
ERROR 1062 (23000): Duplicate entry 'Indian-M1234567' for key 'passengers.uk_passenger_passport'
```

TC-05 …but the same number issued by another country is accepted

```
INSERT INTO passengers (passenger_id, first_name, last_name, email, phone, passport_number, nationality, date_of_birth)
VALUES (4, 'John', 'Smith', 'john.smith@example.com', '+447700900123', 'M1234567', 'British', '1985-03-03');
SELECT passenger_id, first_name, passport_number, nationality FROM passengers WHERE passport_number = 'M1234567';
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO passengers (passenger_id, first_name, last_name, email, phone, passport_number, nationality
, date_of_birth)
    -> VALUES (4, 'John', 'Smith', 'john.smith@example.com', '+447700900123', 'M1234567', 'British', '1985-03-
03');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT passenger_id, first_name, passport_number, nationality FROM passengers WHERE passport_number = '
M1234567';
+--------------+------------+-----------------+-------------+
| passenger_id | first_name | passport_number | nationality |
+--------------+------------+-----------------+-------------+
|            1 | Test       | M1234567        | Indian      |
|            4 | John       | M1234567        | British     |
+--------------+------------+-----------------+-------------+
2 rows in set (0.00 sec)
```

TC-06 A flight number identifies one route

```
INSERT INTO routes (flight_number, origin_code, destination_code, scheduled_departure, duration_min, distance_km)
VALUES ('TS101', 'DEL', 'HYD', '10:00', 135, 1265);
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO routes (flight_number, origin_code, destination_code, scheduled_departure, duration_min, di
stance_km)
    -> VALUES ('TS101', 'DEL', 'HYD', '10:00', 135, 1265);
ERROR 1062 (23000): Duplicate entry 'TS101' for key 'routes.uk_route_flight_number'
```

TC-07 A route operates at most once per day

```
INSERT INTO flights (route_id, aircraft_id, departure_time, arrival_time, base_price)
VALUES (1, 1, '2026-12-01 18:00', '2026-12-01 20:10', 4500);
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO flights (route_id, aircraft_id, departure_time, arrival_time, base_price)
    -> VALUES (1, 1, '2026-12-01 18:00', '2026-12-01 20:10', 4500);
ERROR 1062 (23000): Duplicate entry '1-2026-12-01' for key 'flights.uk_flight_route_day'
```

TC-08 CHECK: airport codes are three capital letters

```
INSERT INTO airports (airport_code, airport_name, city) VALUES ('hy1', 'Bad Code', 'Nowhere');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO airports (airport_code, airport_name, city) VALUES ('hy1', 'Bad Code', 'Nowhere');
ERROR 3819 (HY000): Check constraint 'ck_airport_code' is violated.
```

TC-09 CHECK: origin and destination must differ

```
INSERT INTO routes (flight_number, origin_code, destination_code, scheduled_departure, duration_min, distance_km)
VALUES ('TS999', 'HYD', 'HYD', '09:00', 60, 10);
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO routes (flight_number, origin_code, destination_code, scheduled_departure, duration_min, di
stance_km)
    -> VALUES ('TS999', 'HYD', 'HYD', '09:00', 60, 10);
ERROR 3819 (HY000): Check constraint 'ck_route_airports' is violated.
```

TC-10 CHECK: a flight cannot arrive before it departs

```
UPDATE flights SET arrival_time = '2026-12-01 05:00' WHERE flight_id = 1;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE flights SET arrival_time = '2026-12-01 05:00' WHERE flight_id = 1;
ERROR 3819 (HY000): Check constraint 'ck_flight_times' is violated.
```

TC-11 CHECK: passport number format

```
UPDATE passengers SET passport_number = 'AB-12' WHERE passenger_id = 2;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE passengers SET passport_number = 'AB-12' WHERE passenger_id = 2;
ERROR 3819 (HY000): Check constraint 'ck_passenger_passport' is violated.
```

TC-12 ENUM: unknown flight status is rejected

```
UPDATE flights SET flight_status = 'Boarding' WHERE flight_id = 1;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE flights SET flight_status = 'Boarding' WHERE flight_id = 1;
ERROR 1265 (01000): Data truncated for column 'flight_status' at row 1
```

TC-13 FK: a ticket must refer to an existing flight

```
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number, fare)
VALUES (1, 99, 1, '10A', 5000);
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number, fare)
    -> VALUES (1, 99, 1, '10A', 5000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`airline_reservation`.`t
icket_details`, CONSTRAINT `fk_ticket_flight` FOREIGN KEY (`flight_id`) REFERENCES `flights` (`flight_id`) ON
DELETE RESTRICT ON UPDATE CASCADE)
```

TC-14 Sell a seat giving only booking, flight, traveller and seat

```
INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 1, 1, '12A');
SELECT ticket_id, passenger_id, class_code, seat_number, fare, ticket_status, active_seat
FROM ticket_details;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 1, 1, '12A');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT ticket_id, passenger_id, class_code, seat_number, fare, ticket_status, active_seat
    -> FROM ticket_details;
+-----------+--------------+------------+-------------+---------+---------------+-------------+
| ticket_id | passenger_id | class_code | seat_number | fare    | ticket_status | active_seat |
+-----------+--------------+------------+-------------+---------+---------------+-------------+
|         1 |            1 | EC         | 12A         | 5000.00 | Booked        | 12A         |
+-----------+--------------+------------+-------------+---------+---------------+-------------+
1 row in set (0.00 sec)
```

TC-15 CHECK: seat numbers look like 12A

```
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 2, 'Z9');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 2, 'Z9');
ERROR 3819 (HY000): Check constraint 'ck_ticket_seat' is violated.
```

TC-16 No double-booking: seat 12A is already taken

```
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 2, '12A');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 2, '12A');
ERROR 1062 (23000): Duplicate entry '1-12A' for key 'ticket_details.uk_ticket_seat'
```

TC-17 A traveller cannot hold two seats on the same flight

```
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 1, '12B');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 1, '12B');
ERROR 1062 (23000): Duplicate entry '1-1' for key 'ticket_details.uk_ticket_traveller'
```

TC-18 Cabin capacity: the second economy seat sells, the third is refused

```
INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
VALUES (2, 1, 1, 2, '12B');
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 3, '12C');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (2, 1, 1, 2, '12B');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 3, '12C');
ERROR 1644 (45000): Ticket refused: no seats left in this travel class
```

TC-19 Cancelling a ticket releases its seat and capacity

```
UPDATE ticket_details SET ticket_status = 'Cancelled' WHERE ticket_id = 2;
INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
VALUES (3, 1, 1, 3, '12B');
SELECT ticket_id, passenger_id, seat_number, ticket_status, active_seat FROM ticket_details;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE ticket_details SET ticket_status = 'Cancelled' WHERE ticket_id = 2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (3, 1, 1, 3, '12B');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT ticket_id, passenger_id, seat_number, ticket_status, active_seat FROM ticket_details;
+-----------+--------------+-------------+---------------+-------------+
| ticket_id | passenger_id | seat_number | ticket_status | active_seat |
+-----------+--------------+-------------+---------------+-------------+
|         1 |            1 | 12A         | Booked        | 12A         |
|         2 |            2 | 12B         | Cancelled     | NULL        |
|         3 |            3 | 12B         | Booked        | 12B         |
+-----------+--------------+-------------+---------------+-------------+
3 rows in set (0.00 sec)
```

TC-20 Business fare is computed as base price × 3.20 when no fare is given

```
INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, class_code, seat_number)
VALUES (4, 1, 1, 2, 'BU', '1A');
SELECT ticket_id, passenger_id, class_code, seat_number, fare FROM ticket_details WHERE ticket_id = 4;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (ticket_id, booking_id, flight_id, passenger_id, class_code, seat_number)
    -> VALUES (4, 1, 1, 2, 'BU', '1A');
Query OK, 1 row affected (0.00 sec)

mysql> SELECT ticket_id, passenger_id, class_code, seat_number, fare FROM ticket_details WHERE ticket_id = 4;
+-----------+--------------+------------+-------------+----------+
| ticket_id | passenger_id | class_code | seat_number | fare     |
+-----------+--------------+------------+-------------+----------+
|         4 |            2 | BU         | 1A          | 16000.00 |
+-----------+--------------+------------+-------------+----------+
1 row in set (0.00 sec)
```

TC-21 A failed attempt and one success are allowed; a second success is not

```
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
VALUES (1, 26000, 'Credit Card', 'Failed',  'TXN0001'),
       (1, 26000, 'UPI',         'Success', 'TXN0002');
SELECT payment_id, amount, payment_method, payment_status, transaction_ref, success_booking FROM payments;
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
VALUES (1, 26000, 'UPI', 'Success', 'TXN0003');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
    -> VALUES (1, 26000, 'Credit Card', 'Failed',  'TXN0001'),
    ->        (1, 26000, 'UPI',         'Success', 'TXN0002');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT payment_id, amount, payment_method, payment_status, transaction_ref, success_booking FROM paymen
ts;
+------------+----------+----------------+----------------+-----------------+-----------------+
| payment_id | amount   | payment_method | payment_status | transaction_ref | success_booking |
+------------+----------+----------------+----------------+-----------------+-----------------+
|          1 | 26000.00 | Credit Card    | Failed         | TXN0001         |            NULL |
|          2 | 26000.00 | UPI            | Success        | TXN0002         |               1 |
+------------+----------+----------------+----------------+-----------------+-----------------+
2 rows in set (0.00 sec)

mysql> INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
    -> VALUES (1, 26000, 'UPI', 'Success', 'TXN0003');
ERROR 1062 (23000): Duplicate entry '1' for key 'payments.uk_one_successful_payment'
```

TC-22 CHECK: a payment amount must be positive

```sql
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
VALUES (1, 0, 'UPI', 'Failed', 'TXN0004');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
    -> VALUES (1, 0, 'UPI', 'Failed', 'TXN0004');
ERROR 3819 (HY000): Check constraint 'ck_payment_amount' is violated.
```

TC-23 Gateway transaction references are unique

```sql
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
VALUES (1, 100, 'UPI', 'Failed', 'TXN0001');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
    -> VALUES (1, 100, 'UPI', 'Failed', 'TXN0001');
ERROR 1062 (23000): Duplicate entry 'TXN0001' for key 'payments.uk_payment_txn_ref'
```

TC-24 Cancelling the flight cancels every live ticket on it

```sql
UPDATE flights SET flight_status = 'Cancelled' WHERE flight_id = 1;
SELECT ticket_id, passenger_id, seat_number, ticket_status FROM ticket_details;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE flights SET flight_status = 'Cancelled' WHERE flight_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT ticket_id, passenger_id, seat_number, ticket_status FROM ticket_details;
+-----------+--------------+-------------+---------------+
| ticket_id | passenger_id | seat_number | ticket_status |
+-----------+--------------+-------------+---------------+
|         1 |            1 | 12A         | Cancelled     |
|         2 |            2 | 12B         | Cancelled     |
|         3 |            3 | 12B         | Cancelled     |
|         4 |            2 | 1A          | Cancelled     |
+-----------+--------------+-------------+---------------+
4 rows in set (0.00 sec)
```

TC-25 A cancelled flight is no longer open for booking

```
INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
VALUES (1, 1, 3, '14C');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO ticket_details (booking_id, flight_id, passenger_id, seat_number)
    -> VALUES (1, 1, 3, '14C');
ERROR 1644 (45000): Ticket refused: flight is no longer open for booking
```

TC-26 A cancelled booking cannot be paid

```
INSERT INTO bookings (booking_id, passenger_id, booking_reference, status) VALUES (2, 2, 'TEST02', 'Cancelled');
INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
VALUES (2, 5000, 'UPI', 'Success', 'TXN0005');
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> INSERT INTO bookings (booking_id, passenger_id, booking_reference, status) VALUES (2, 2, 'TEST02', 'Can
celled');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO payments (booking_id, amount, payment_method, payment_status, transaction_ref)
    -> VALUES (2, 5000, 'UPI', 'Success', 'TXN0005');
ERROR 1644 (45000): Payment refused: booking is cancelled
```

TC-27 RESTRICT: an airport used by a route cannot be deleted

```
DELETE FROM airports WHERE airport_code = 'HYD';
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DELETE FROM airports WHERE airport_code = 'HYD';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`airline_reservation
`.`routes`, CONSTRAINT `fk_route_origin` FOREIGN KEY (`origin_code`) REFERENCES `airports` (`airport_code`) ON
 DELETE RESTRICT ON UPDATE RESTRICT)
```

TC-28 RESTRICT on update: an IATA code in use cannot be changed

```
UPDATE airports SET airport_code = 'HDD' WHERE airport_code = 'HYD';
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> UPDATE airports SET airport_code = 'HDD' WHERE airport_code = 'HYD';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`airline_reservation
`.`routes`, CONSTRAINT `fk_route_origin` FOREIGN KEY (`origin_code`) REFERENCES `airports` (`airport_code`) ON
 DELETE RESTRICT ON UPDATE RESTRICT)
```

TC-29 RESTRICT: a passenger with bookings cannot be deleted

```
DELETE FROM passengers WHERE passenger_id = 1;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DELETE FROM passengers WHERE passenger_id = 1;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`airline_reservation
`.`bookings`, CONSTRAINT `fk_booking_passenger` FOREIGN KEY (`passenger_id`) REFERENCES `passengers` (`passeng
er_id`) ON DELETE RESTRICT ON UPDATE CASCADE)
```

TC-30 CASCADE: deleting a booking deletes its tickets and payments (rolled back)

```
START TRANSACTION;
DELETE FROM bookings WHERE booking_id = 1;
SELECT (SELECT COUNT(*) FROM ticket_details) AS tickets_left, (SELECT COUNT(*) FROM payments) AS payments_left;
ROLLBACK;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> DELETE FROM bookings WHERE booking_id = 1;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT (SELECT COUNT(*) FROM ticket_details) AS tickets_left, (SELECT COUNT(*) FROM payments) AS paymen
ts_left;
+--------------+---------------+
| tickets_left | payments_left |
+--------------+---------------+
|            0 |             0 |
+--------------+---------------+
1 row in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

TC-31 ON UPDATE CASCADE: renaming a class code updates its tickets (rolled back)

```
START TRANSACTION;
UPDATE travel_classes SET class_code = 'BZ' WHERE class_code = 'BU';
SELECT ticket_id, class_code FROM ticket_details WHERE ticket_id = 4;
ROLLBACK;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE travel_classes SET class_code = 'BZ' WHERE class_code = 'BU';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT ticket_id, class_code FROM ticket_details WHERE ticket_id = 4;
+-----------+------------+
| ticket_id | class_code |
+-----------+------------+
|         4 | BZ         |
+-----------+------------+
1 row in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

8.3 Removing the fixture

```
DELETE FROM payments;   DELETE FROM ticket_details; DELETE FROM bookings;
DELETE FROM passengers; DELETE FROM flights;        DELETE FROM routes;
DELETE FROM aircraft;   DELETE FROM travel_classes; DELETE FROM airports;
SELECT (SELECT COUNT(*) FROM flights) AS flights, (SELECT COUNT(*) FROM ticket_details) AS tickets,
       (SELECT COUNT(*) FROM passengers) AS passengers;
```

MySQL 8.0 Command Line Client — airline_reservation

```
mysql> DELETE FROM payments;
Query OK, 2 rows affected (0.00 sec)

mysql> DELETE FROM ticket_details;
Query OK, 4 rows affected (0.00 sec)

mysql> DELETE FROM bookings;
Query OK, 2 rows affected (0.00 sec)

mysql> DELETE FROM passengers;
Query OK, 4 rows affected (0.00 sec)

mysql> DELETE FROM flights;
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM routes;
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM aircraft;
Query OK, 1 row affected (0.00 sec)

mysql> DELETE FROM travel_classes;
Query OK, 2 rows affected (0.00 sec)

mysql> DELETE FROM airports;
Query OK, 2 rows affected (0.00 sec)

mysql> SELECT (SELECT COUNT(*) FROM flights) AS flights, (SELECT COUNT(*) FROM ticket_details) AS tickets,
    ->        (SELECT COUNT(*) FROM passengers) AS passengers;
+---------+---------+------------+
| flights | tickets | passengers |
+---------+---------+------------+
|       0 |       0 |          0 |
+---------+---------+------------+
1 row in set (0.00 sec)
```

9. Summary of Test Results

Test	Type	Operation	Expected	Actual
TC-01	Entity	Primary key: an airport code can exist only once	Error	**Error 1062** airports.PRIMARY
TC-02	Entity	NOT NULL: e-mail is mandatory	Error	**Error 1364** NOT NULL
TC-03	Unique	Two passengers cannot share an e-mail	Error	**Error 1062** passengers.uk_passenger_email
TC-04	Unique	Same passport number and same nationality is rejected	Error	**Error 1062** passengers.uk_passenger_passport
TC-05	Unique	…but the same number issued by another country is accepted	Success	**Success**
TC-06	Unique	A flight number identifies one route	Error	**Error 1062** routes.uk_route_flight_number
TC-07	Unique (generated)	A route operates at most once per day	Error	**Error 1062** flights.uk_flight_route_day
TC-08	Domain	CHECK: airport codes are three capital letters	Error	**Error 3819** ck_airport_code
TC-09	Domain	CHECK: origin and destination must differ	Error	**Error 3819** ck_route_airports
TC-10	Domain	CHECK: a flight cannot arrive before it departs	Error	**Error 3819** ck_flight_times
TC-11	Domain	CHECK: passport number format	Error	**Error 3819** ck_passenger_passport
TC-12	Domain	ENUM: unknown flight status is rejected	Error	**Error 1265** ENUM
TC-13	Referential	FK: a ticket must refer to an existing flight	Error	**Error 1452** FK
TC-14	Default + trigger	Sell a seat giving only booking, flight, traveller and seat	Success	**Success**

Test	Type	Operation	Expected	Actual
TC-15	Domain	CHECK: seat numbers look like 12A	Error	**Error 3819** ck_ticket_seat
TC-16	Unique (generated)	No double-booking: seat 12A is already taken	Error	**Error 1062** ticket_details.uk_ticket_seat
TC-17	Unique (generated)	A traveller cannot hold two seats on the same flight	Error	**Error 1062** ticket_details.uk_ticket_traveller
TC-18	Trigger	Cabin capacity: the second economy seat sells, the third is refused	Error	**Error 1644** Ticket refused: no seats left in this travel class
TC-19	Accepted	Cancelling a ticket releases its seat and capacity	Success	**Success**
TC-20	Trigger	Business fare is computed as base price × 3.20 when no fare is given	Success	**Success**
TC-21	Unique (generated)	A failed attempt and one success are allowed; a second success is not	Error	**Error 1062** payments.uk_one_successful_payment
TC-22	Domain	CHECK: a payment amount must be positive	Error	**Error 3819** ck_payment_amount
TC-23	Unique	Gateway transaction references are unique	Error	**Error 1062** payments.uk_payment_txn_ref
TC-24	Trigger	Cancelling the flight cancels every live ticket on it	Success	**Success**
TC-25	Trigger	A cancelled flight is no longer open for booking	Error	**Error 1644** Ticket refused: flight is no longer open for booking
TC-26	Trigger	A cancelled booking cannot be paid	Error	**Error 1644** Payment refused: booking is cancelled
TC-27	Referential	RESTRICT: an airport used by a route cannot be deleted	Error	**Error 1451** FK
TC-28	Referential	RESTRICT on update: an IATA code in use cannot be changed	Error	**Error 1451** FK
TC-29	Referential	RESTRICT: a passenger with bookings cannot be deleted	Error	**Error 1451** FK
TC-30	Referential	CASCADE: deleting a booking deletes its tickets and payments (rolled back)	Success	**Success**
TC-31	Referential	ON UPDATE CASCADE: renaming a class code updates its tickets (rolled back)	Success	**Success**

All 31 test cases produced the expected outcome. Error codes: 1062 duplicate key, 1364 missing mandatory value, 3819 CHECK violated, 1265 invalid ENUM value, 1452 parent row missing, 1451 parent row still referenced, 1644 rule raised by a trigger.

Result

The airline database now enforces its own business rules. Keys keep every reference valid, generated-column UNIQUE keys make double-booked seats and double payments impossible, CHECK constraints keep codes, times and amounts realistic, and the triggers keep seat inventory and flight cancellations consistent. The test cases demonstrate every rule.